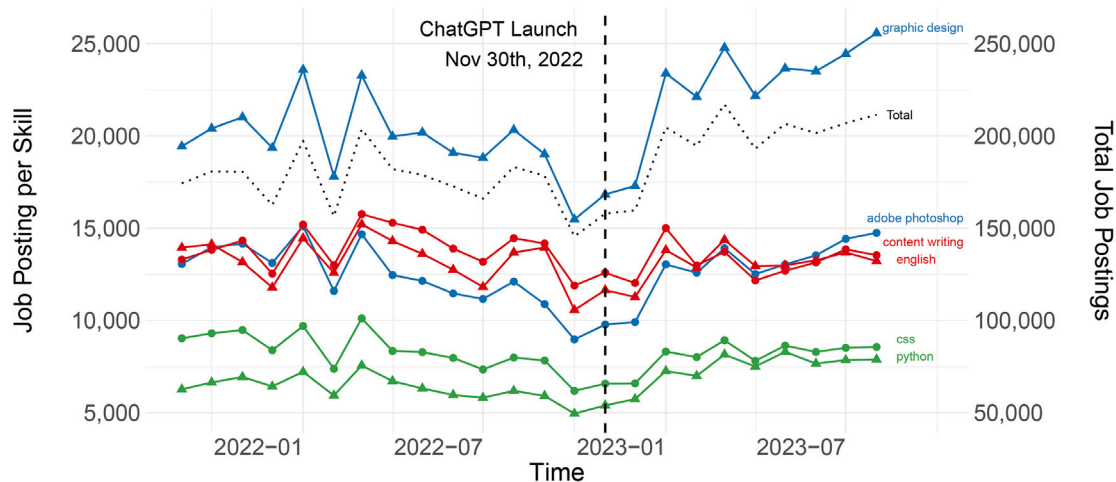




(a) Skill clusters in the embedding space



(b) Wordclouds of most frequent, important words per category



(c) Development of skill-specific job demand over time

Fig. 3. (a) Two-dimensional representation of the skill embedding space with 1% random sample of jobs plotted and the three most frequent skill clusters per AI exposure category highlighted. (b) Wordclouds of the 30 most observed important skills per category. (c) Development of job postings between September 2021 and August 2023 over time for important skills in each AI exposure category.

Table 2

Difference in Difference regression results - models (1)–(4) show the overall effects, models (5)–(8) the effects by project length, and models (9)–(11) show the results by worker experience level. Overall, job demand in the substitutable categories decreased compared to the counterfactual.

	Overall effects				Effects by project length				Effects by worker experience		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Intercept	5.21*** (0.19)	7.21*** (0.70)			3.95*** (0.18)	3.20*** (0.19)	2.49*** (0.19)	3.09*** (0.22)	2.96*** (0.21)	4.65*** (0.20)	4.01*** (0.19)
Substitutable	0.05 (0.37)	−0.13 (0.34)	0.05 (0.38)		0.02 (0.36)	−0.21 (0.37)	−0.21 (0.39)	−0.44 (0.41)	0.25 (0.43)	0.04 (0.37)	−0.01 (0.37)
Complemented	−0.10 (0.24)	−0.05 (0.21)	−0.10 (0.24)		−0.08 (0.22)	0.01 (0.25)	−0.16 (0.24)	−0.21 (0.27)	−0.21 (0.26)	−0.13 (0.24)	−0.05 (0.24)
After	0.16*** (0.04)	0.12** (0.05)			0.11** (0.04)	0.14*** (0.04)	0.19*** (0.04)	0.06 (0.05)	0.29*** (0.05)	0.13*** (0.04)	0.17*** (0.04)
Substitutable × After	−0.28** (0.09)	−0.29** (0.09)	−0.28** (0.09)	−0.28** (0.09)	−0.29** (0.09)	−0.14 (0.10)	−0.12 (0.07)	−0.06 (0.09)	−0.19* (0.08)	−0.28** (0.09)	−0.30** (0.10)
Complemented × After	−0.05 (0.05)	−0.01 (0.05)	−0.05 (0.05)	−0.05 (0.05)	−0.02 (0.05)	0.01 (0.05)	−0.02 (0.06)	−0.06 (0.07)	−0.16* (0.07)	−0.03 (0.05)	−0.04 (0.05)
Controls	–	✓	–	–	Exp. project duration (x weeks or less)				Worker experience level		
Week fixed effects	–	–	✓	✓	3	9	18	52	Novice	Intermediate	Veteran
Cluster fixed effects	–	–	–	✓							
Observations	11 960	11 524	11 960	11 960	11 955	11 847	11 443	11 615	11 552	11 960	11 942
R ²	0.005	0.13	0.02	0.97	0.004	0.007	0.01	0.01	0.02	0.005	0.004

Notes: *p < 0.05, **p < 0.01, ***p < 0.001.

3.2. Regression results

Here, we turn to investigating the effect the introduction of ChatGPT had on job demand in the three different AI exposure categories: substitutable, complementary, and unaffected. In Fig. 4(a), we plot the total number of job postings in each AI exposure category. All three categories follow approximately similar time trends before the introduction of ChatGPT. This is confirmed in the event study plot shown in Fig. 4(b). It illustrates that the difference between the two treatment labels (complementary and substitutable) and the unaffected control group remained on average constant before the launch of ChatGPT.

Specifically, no significant differences in time trends between the treatment and control groups are evident, and the disparity between both treatment groups and the control group is effectively negligible across all but one week in the pre-treatment period.²

We present our difference-in-differences regression results in Table 2 (models 1–4). Model (1) shows the results without any fixed effects. In model (2), we add two control variables: the percentages of projects with different durations (under 3 weeks, 3–9 weeks, 9–18 weeks, and 18–52 weeks) and projects aimed at three experience levels of freelancers (novice, intermediate, and veteran). Since our unit of observation is the number of new job postings within each cluster-by-week cell, we introduce these control variables as percentage shares of the number of job postings within each cell. In model (3), we introduce week fixed effects, and in model (4), we further incorporate cluster fixed effects.

Overall, we find the following: After the launch of ChatGPT, the number of job postings in the substitutable clusters decreased by 24% ($100 * e^{-0.28} - 1$), relative to the unaffected clusters.³

The decrease can be decomposed into two effects: (i) the number of new job postings in the unaffected clusters increased by 17% after the introduction of ChatGPT ($100 * e^{-0.16} - 1$) and (ii) the number of substitutable job postings decreased by 7% ($-24\% + 17\%$) after the introduction of GPT.

Additionally, we find that the point estimates on the term Complementary×After are also negative but not statistically significant. As we show below, this is partly due to the fact that some complementary clusters experienced a decrease in demand after the launch of ChatGPT while others experienced an increase.

The difference-in-differences estimates remain highly consistent across the four different model specifications, increasing our confidence that no unobserved confounders are driving our results.

3.2.1. Project duration heterogeneity

To investigate the heterogeneity of the effects of ChatGPT by expected project length, we split the data into subsets based on the expected project length and then aggregate the four data subsets into week × cluster cells. The results are presented in Table 2 (models 5–8). We find that the demand reduction in the substitutable clusters is driven by a decrease in short projects lasting three weeks or less.

Conversely, we observe an increase in labor demand in the unaffected clusters even for short projects, as evidenced by the positive coefficient on the After indicator variable. The difference between the unaffected and complemented clusters is not statistically

² An exception is noted during the week of the 31st of January 2022 when a transitory spike occurred in the substitution group relative to the two other groups. We cannot ascertain whether this is related to an error in the data collection or an exceptional number of job postings on the platform. Nevertheless, as detailed in Appendix E, our findings remain substantially unchanged even when we exclude the data from this anomalous week in our pre-treatment analysis.

³ Throughout this section, we transform log points into percentages using the formula $100 * (e^x - 1)$, i.e., $100 * (e^{-0.28} - 1) \approx -24\%$.

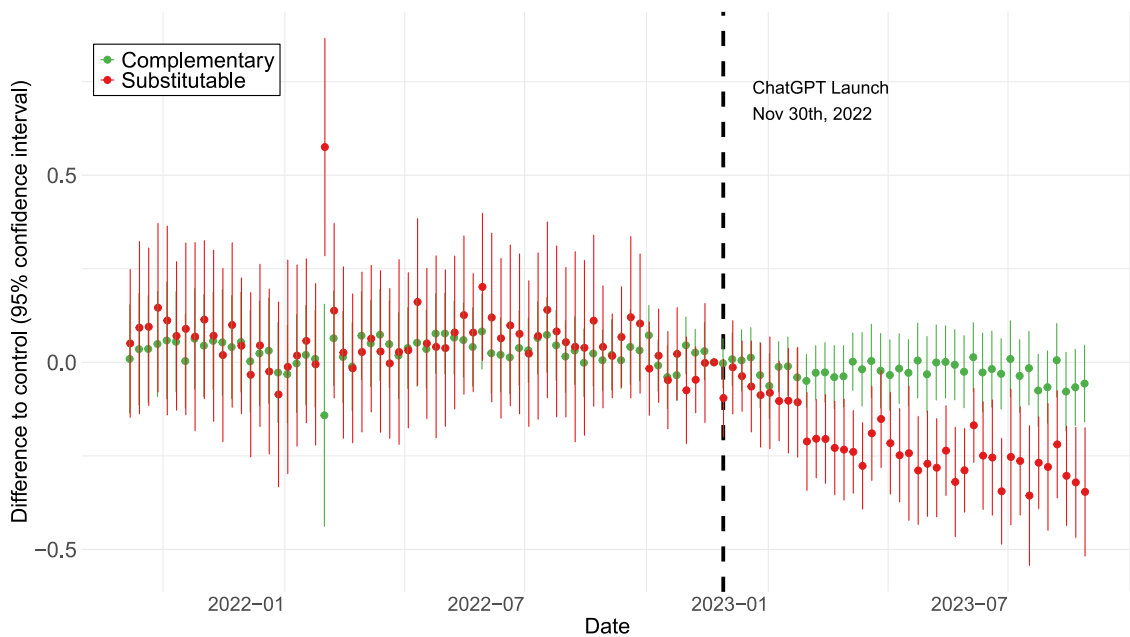
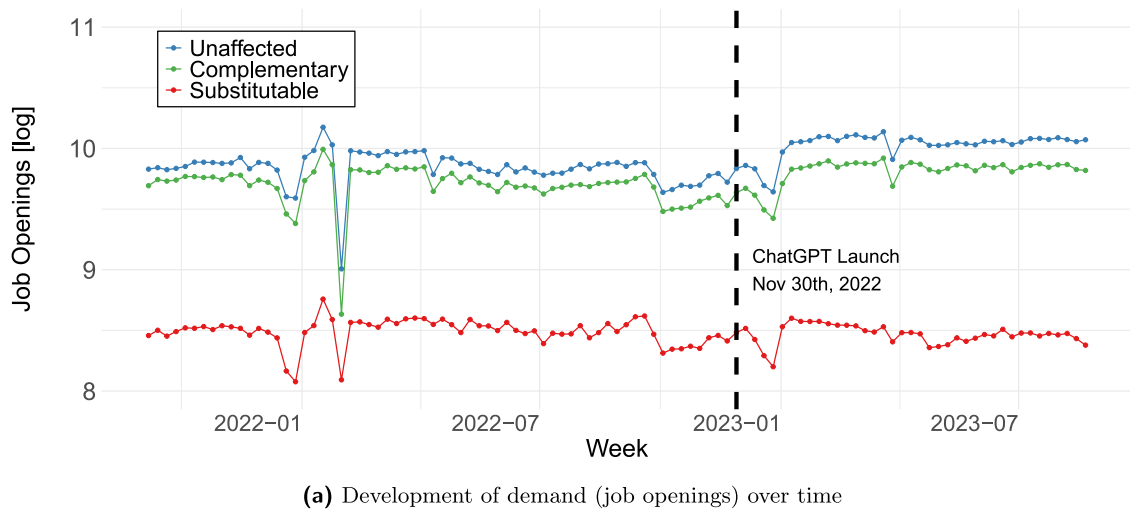


Fig. 4. (a) Logarithm of aggregated demand per AI exposure category between September 2021 and August 2023. (b) OLS coefficient estimates and their 95% confidence intervals showing the difference between each treatment group and the control.

significant and is also much smaller economically across all project duration categories. These findings indicate that the launch of ChatGPT resulted in a shift of work from short gigs to longer projects in the substitutable clusters.

3.2.2. Worker experience level heterogeneity

To study the heterogeneous effects with respect to worker experience levels, we split our data into three subsets based on the employers' desired worker experience and then aggregate the data into week $\times$ cluster cells (analogous to the previous subsection). We present our results in Table 2 (models 9–11).

The effect of the ChatGPT launch on the demand for substitutable skills is roughly similar across desired worker experience levels. The regression coefficient on the Substitutable $\times$ After term is slightly smaller in absolute value in the novice subset compared to the other two subsets. However, due to large standard errors, this difference is not statistically significant. More interestingly,

we find a negative and statistically significant difference in differences estimate for the complementary group for novice workers. This indicates that the demand for novice labor decreased in the complemented cluster after the launch of ChatGPT. This finding highlights the importance of considering other dimensions of heterogeneity beyond skill requirements when studying the effects of LLMs on the labor market.

3.2.3. Effect heterogeneity by skill cluster

Next, we formally examine the heterogeneity of the impact of the ChatGPT launch across clusters in the substitutable and complementary AI exposure categories by estimating the regression in Eq. (2) presented in the methods section.

Fig. 5 illustrates the estimated changes in demand for each skill cluster in the substitutable and complementary category. Most substitutable clusters exhibit a decline compared to the counterfactual: Writing related to real estate and 'about us' pages show the most substantial decreases, at 52% and 59%, respectively. The largest cluster in our sample of affected skill clusters—content writing for blogs—declined by 20%. Translation is also significantly impacted, with the 'Western European Languages' category experiencing a 23% decline. In the complementary clusters, demand changes are mixed, with some clusters showing significant decreases and others increases. Notably, the demand for jobs in the cluster 'AI-powered chatbots' nearly tripled, with a 179% increase. The cluster of 'Machine learning' also shows a significant rise, with demand increasing by 29%.

Cluster-specific estimates should be interpreted with caution for two reasons. First, while the complementary and substitutable clusters show no statistically significant pre-trends on average, individual clusters may still display notable pre-trends, which could influence some results. Rather than excluding clusters with statistically significant pre-trends, we present results for all clusters. Given the large number of comparisons, some statistically significant pre-trends will occur by chance, and it is difficult to distinguish between true pre-trends and those arising randomly. Second, comparing estimates requires adjusting for multiple hypothesis testing. However, adjusting for every pairwise comparison is impractical, as it would inflate standard errors and make Fig. 5 uninformative.⁴ We therefore recommend focusing on the top and bottom of the distribution shown in Fig. 5.

Notwithstanding these caveats, the main takeaway from Fig. 5 is that ChatGPT resulted in substantial demand increases for clusters related to AI for chatbot development, lead generation, and machine learning and significantly decreased demand for writing and translation jobs.⁵

The results from Fig. 5 align with results on the performance of LLMs across languages (Li et al., 2024) and domains (del Rio-Chanona et al., 2024). The significant decline in demand for translation of Western European languages compared to less common languages, such as Arabic and Hebrew, is consistent with the superior performance of LLMs in Western languages, making them more easily substitutable. Clusters related to writing that do not exhibit a significant decrease in demand include Spanish localization and translation, fiction writing, editing, and script writing for explainer videos. These skills require an understanding of language nuances, creativity, and interaction with video content, which explains their relatively stable demand. In programming, the most affected clusters are related to JavaScript, HTML, and CSS, which are also among the most affected programming languages in collaborative programming platforms such as Stack Overflow (del Rio-Chanona et al., 2024). The increase in demand for AI-powered chatbot development and machine learning is intuitive, as businesses increasingly adopt these technologies.

3.2.4. Sensitivity analyses

We have conducted a series of sensitivity analyses to demonstrate that our results are robust to reasonable variations in the regression model and pass standard checks for identification. While the main exhibits related to these exercises are provided in the appendices, we briefly discuss each of them here individually.

Placebo tests. We first provide a standard 'placebo treatment time' test to show that our results do not capture underlying trends that are unrelated to ChatGPT. We do this by entirely dropping the post-ChatGPT period from our data. Thereafter, we introduce a placebo treatment set to 30th May 2022. In Appendix F, we show that the difference in differences estimates are indistinguishable from zero in this case.

Dropping graphic design. So far, we have focused on studying the effects of LLMs on the labor market, ignoring other breakthroughs in generative artificial intelligence technologies related to image generation (such as Dall-E, originally released in 2021, and Midjourney, originally released in 2022). According to our labeling which only focuses on text-based models, graphic design tasks are mostly in unaffected clusters. Nonetheless, one could argue that our results might be affected by graphic design jobs in the control group, which were likely affected by the advent of advanced AI image generation capabilities. We addressed this by dropping all clusters that include 'graphic design' as one of their important skills from the control group. We report the results of this robustness check in Appendix G, which are virtually unchanged.

⁴ Depending on the exact method for multiple hypothesis testing correction, we would need to inflate the standard errors by some constant for every 70×69 pairwise comparison we make.

⁵ The standard errors on cluster-specific estimates are very similar because the analysis sample consists of week-by-cluster cells, resulting in identical sample sizes across clusters.